

- Neuroleptic malignant syndrome (NMS) (has not been observed in patients taking deutetrabenazine, but has been observed in patients taking tetrabenazine)
- Risk of depression and suicidal thoughts and behavior in patients with Huntington's disease

Weight Gain



- Reported but not expected

Sedation



- Many experience and/or can be significant in amount

What to Do About Side Effects

- Wait
- Wait
- Wait
- Reduce dose or discontinue if agitation, akathisia, restlessness, or parkinsonism occurs

Best Augmenting Agents for Side Effects

Deutetrabenazine is itself an augmenting agent to treat a side effect of antipsychotic drugs

Dosing and use

Usual Dosage Range

- Tardive dyskinesia: 12–48 mg/day in two divided doses
- Chorea associated with Huntington's disease: 6–48 mg/day (in two divided doses for total daily dosages of 12 mg or more)

Dosage Forms

- Tablet 6 mg, 9 mg, 12 mg

How to Dose

- Tardive dyskinesia: initial dose 12 mg in two divided doses; titrate by 6 mg/day in weekly intervals; maximum recommended dose is 48 mg/day in two divided doses
- Chorea associated with Huntington's disease: initial dose 6 mg once daily; titrate by 6 mg/day in weekly intervals; doses above 12 mg/day should be administered in two divided doses; maximum recommended dose is 48 mg/day in two divided doses



Dosing Tips

Should be taken with food because that is how clinical trials were conducted; however, food has no effect on the AUC of deutetrabenazine's active metabolites, but increased C_{max} by 50%

Tablet should be swallowed whole and should not be chewed or crushed

If switching from tetrabenazine: discontinue tetrabenazine and initiate deutetrabenazine the next day:

Current tetrabenazine daily dose	Initial deutetrabenazine dose
12.5 mg	6 mg once daily
25 mg	6 mg twice daily
37.5 mg	9 mg twice daily
50 mg	12 mg twice daily
62.5 mg	15 mg twice daily
75 mg	18 mg twice daily
87.5 mg	21 mg twice daily
100 mg	24 mg twice daily

If switching from reserpine: at least 20 days should elapse after stopping reserpine before starting deutetrabenazine

Overdose

- Limited experience
- Overdose with tetrabenazine: acute dystonia, oculogyric crisis, nausea and vomiting, sweating, sedation, hypotension, confusion, diarrhea, hallucinations, rubor, and tremor have been reported

Long-Term Use

- Long-term clinical trials have not been conducted

Habit Forming

- No

How to Stop

- Taper not necessary

Pharmacokinetics

- Deutetrabenazine is a deuterated form of tetrabenazine
- Deuteration is a process in which the stable isotope deuterium replaces selected hydrogen atoms, resulting in a compound with similar pharmacodynamic effects but different pharmacokinetic profiles, because the carbon–deuterium covalent bond requires 8 times more energy to break than a carbon–hydrogen bond
- Substitution of deuterium for hydrogen slows the breakdown of metabolites; thus, the active metabolites of deutetrabenazine have longer half-lives and greater absorption than those of tetrabenazine
- Deutetrabenazine is broken down into 4 metabolites
- Half-life of total (a+b)-HTBZ from deutetrabenazine is approximately 9 to 10 hours
- Metabolized by CYP450 2D6, but deuteration means that, unlike tetrabenazine, CYP450 2D6 genotyping is not necessary

Drug Interactions

Strong CYP450 2D6 inhibitors (e.g., paroxetine, fluoxetine) may increase exposure to deutetrabenazine; maximum recommended dose in patients taking strong CYP450 2D6 inhibitors or who are known CYP2D6 poor metabolizers is 36 mg/day

May be additive sedative effects if deutetrabenazine is given concomitantly with alcohol or other sedating drugs

Other Warnings/Precautions

May cause an increase in QT interval, although the degree of QT prolongation is not clinically significant at concentrations expected with recommended dosing

Avoid use in patients with congenital long QT syndrome or with arrhythmias associated with a prolonged QT interval

For patients at increased risk of prolonged QT interval, assess the QT interval before and after increasing the total dose above 24 mg/day

Due to the increased risk of depression and suicidality in patients with Huntington's disease who take deutetrabenazine, one should monitor for the emergence or worsening of depression, suicidality, or unusual changes in behavior; patients and caregivers should be informed of the risk of depression and suicidality and told to report behaviors of concern promptly

Do Not Use

- In patients with Huntington's disease who are suicidal or have untreated/inadequately treated depression
- In patients with hepatic impairment
- If patient is taking a monoamine oxidase inhibitor (MAOI)
- If patient is taking reserpine, tetrabenazine, or valbenazine
- If there is a proven allergy to deutetrabenazine

Special populations

Renal Impairment

- Not evaluated

Hepatic Impairment

- Contraindicated

Cardiac Impairment

- May cause an increase in QT interval; avoid use in patients with congenital long QT syndrome or with arrhythmias associated with a prolonged QT interval
- For patients at risk for QT prolongation: assess the QT interval before and after increasing the total dosage above 24 mg/day

Elderly

- Some patients may tolerate lower doses better



Children and Adolescents

- Safety and efficacy have not been established



Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- In rat studies, no malformations were observed when deutetrabenazine was administered during the period of organogenesis at doses up to 6 times the maximum recommended human dose
- In rat studies, administration of tetrabenazine during organogenesis through lactation produced an increase in the number of stillborn pups and postnatal pup mortalities

Breast Feeding

- Unknown if deutetrabenazine is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk

The art of psychopharmacology

Potential Advantages

- Multiple doses for flexible dosing
- No CYP450 3A4 drug interactions

Potential Disadvantages

- Expensive
- Requires twice daily dosing (12 mg/day and above)
- Carries a black box warning for depression and suicidality in patients with Huntington's disease

Primary Target Symptoms

- Repetitive involuntary movements of tardive dyskinesia (usually associated with lower facial and distal extremity musculature, e.g., tongue protrusion, writhing of tongue, lip smacking, chewing, blinking, and grimacing)



In clinical trials, deutetrabenazine was well tolerated, with low rates of psychiatric adverse events, including anxiety

Unlike with tetrabenazine, clinical trials with deutetrabenazine did not result in reports of depression or suicidal ideation

The safety profile of deutetrabenazine appears similar regardless of whether a concomitant antipsychotic drug is administered

At therapeutic doses, deutetrabenazine's active metabolites bind predominantly to VMAT2, with low D2 dopamine receptor blockade

Deutetrabenazine's active metabolites have moderate blockade of 5HT7; the clinical significance of this is unknown, although 5HT7 antagonism has been associated with antidepressant and pro-cognitive actions in animal models

Can rarely cause akathisia

Suggested Reading

Anderson KE , Stamler D , Davis MD , et al. Deutetrabenazine for treatment of involuntary movements in patients with tardive dyskinesia (AIM-TD): a double-blind, randomised, placebo-controlled, phase 3 trial . *Lancet Psychiatry* 2017 ;**4** (8):595 –604 .

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Fernandez HH , Factor SA , Hauser RA , et al. Randomized controlled trial of deutetrabenazine for tardive dyskinesia: the ARM-TD study . *Neurology* 2017 ;**88** (21):2003 –10 .

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Frank S , Stamler D , Kayson E , et al. Safety of converting from tetrabenazine to deutetrabenazine for the treatment of chorea . *JAMA Neurol* 2017 ;**74** (8):977 –82 .

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Huntington Study Group, Frank S , Testa CM , et al. Effect of
deutetrabenazine on chorea among patients with Huntington disease: a
randomized clinical trial. *JAMA* 2016 ;**316** (1):40 –50 .
.....

Meyer JM . Forgotten but not gone: new developments in the understanding
and treatment of tardive dyskinesia . *CNS Spectr* 2016 ;**21** (S1):13 –24 .

Dextromethorphan

Therapeutics

Brands

- Nuedexta (in combination with quinidine)
- *see index for additional brand names*

Generic?

No

Class

- Noncompetitive N-methyl-D-aspartate (NMDA) receptor antagonist and sigma 1 agonist

Commonly Prescribed for

(bold for FDA approved)

- **Pseudobulbar affect (PBA)**
- Diabetic peripheral neuropathic pain
- Unstable mood and affect in PTSD and mild traumatic brain injury
- Third-line for treatment-resistant depression

How the Drug Works

- Dextromethorphan reduces glutamate neurotransmission through blocking NMDA receptors and by acting as an agonist at sigma 1 receptors
- Dextromethorphan also has affinity for the serotonin transporter and may therefore modulate serotonin levels
- Quinidine increases availability of dextromethorphan by inhibiting its metabolism via CYP450 2D6

How Long Until It Works

- In clinical trials the rate of pseudobulbar affect episodes was significantly decreased beginning at day 15

If It Works

- Reduces the frequency and severity of episodes of uncontrollable laughing and/or crying

If It Doesn't Work

- Consider switching to a TCA or an SSRI



Best Augmenting Combos for Partial Response or Treatment Resistance

- Often best to attempt another monotherapy prior to resorting to augmentation strategies

Tests

- None for healthy individuals

Side effects

How Drug Causes Side Effects

- Presumably mechanism-related, including central actions at sigma and NMDA receptors causing dissociative symptoms, euphoria, or sedation

Notable Side Effects

- Dizziness, asthenia
- Diarrhea, vomiting
- Cough, peripheral edema
- Urinary tract infection
- Euphoria



Life-Threatening or Dangerous Side Effects

- Immune-mediated thrombocytopenia
- Hepatotoxicity
- Dose-dependent QT prolongation

Weight Gain



- Reported but not expected

Sedation



- Reported but not expected

What to Do About Side Effects

- Wait
- Wait
- Wait
- In a few weeks, switch to another agent or add other drugs

Best Augmenting Agents for Side Effects

- Often best to try another agent prior to resorting to augmentation strategies to treat side effects
- Many side effects are dose-dependent (i.e., they increase as dose increases, or they reemerge until tolerance redevelops)
- Many side effects are time-dependent (i.e., they start immediately upon dosing and upon each dose increase, but go away with time)

Dosing and use

Usual Dosage Range

- 20 mg/10 mg twice per day

Dosage Forms

- Capsule 20 mg/10 mg (dextromethorphan/quinidine)

How to Dose

- Initial 20 mg/10 mg once per day; after 7 days increase to 20 mg/10 mg twice per day



Dosing Tips

- Some patients may tolerate and respond to doses higher than the approved doses, but few controlled studies of high doses

Overdose

- Nausea, dizziness, headache, ventricular arrhythmias, hypotension, coma, respiratory depression, seizures, tachycardia, hyperexcitability, toxic psychosis

Long-Term Use

- Not evaluated

Habit Forming

- No

How to Stop

- Taper not necessary

Pharmacokinetics

- Elimination half-life of dextromethorphan is approximately 13 hours
- Elimination half-life of quinidine is approximately 7 hours
- Dextromethorphan is metabolized by CYP450 2D6, while quinidine inhibits CYP450 2D6
- Quinidine is metabolized by CYP450 3A4



Drug Interactions

- Via CYP450 2D6 inhibition, dextromethorphan/quinidine could increase plasma concentrations of drugs metabolized by CYP450 2D6 (e.g., desipramine), potentially requiring dose reduction of the substrate
- Can cause a fatal “serotonin syndrome” when combined with MAOIs, so do not use with MAOIs or for at least 14 days after MAOIs are stopped
- Do not start an MAOI for at least 5 half-lives (5 to 7 days for most drugs) after discontinuing dextromethorphan/quinidine
- Via P-glycoprotein inhibition, quinidine could increase plasma concentrations of P-glycoprotein substrates such as digoxin, potentially requiring dose reduction of the substrate

- Can theoretically cause serotonin syndrome when combined with serotonin reuptake inhibitors, but not well studied

Other Warnings/Precautions

- Quinidine can cause immune-mediated thrombocytopenia that can be severe or fatal; dextromethorphan/quinidine should be discontinued immediately if thrombocytopenia occurs unless it is clearly not drug-related, and should not be restarted in sensitized patients
- Quinidine has been associated with a lupus-like syndrome involving polyarthritis
- ECG should be monitored if patient must take an agent that prolongs QT interval or that inhibits CYP450 3A4
- Anticholinergic effects of quinidine may lead to worsening in myasthenia gravis and other sensitive conditions

Do Not Use

- If patient is taking an MAOI
- If patient is taking another medication containing quinidine, quinine, or mefloquine
- If patient has a history of quinidine, quinine, or mefloquine-induced thrombocytopenia, hepatitis, or other hypersensitivity reaction
- If patient has prolonged QT interval, congenital long QT syndrome, history suggestive of torsade de pointes, or heart failure

- If patient has complete atrioventricular (AV) block without implanted pacemaker, or patients at high risk of complete AV block
- If patient is taking a drug that prolongs QT interval and is metabolized by CYP450 2D6 (e.g., thioridazine, pimozide)
- If there is a proven allergy to dextromethorphan or quinidine

Special populations

Renal Impairment

- Dose adjustment not necessary in patients with mild to moderate impairment

Hepatic Impairment

- Dose adjustment not necessary in patients with mild to moderate impairment

Cardiac Impairment

- Contraindicated in patients with prolonged QT interval, congenital long QT syndrome, history suggestive of torsade de pointes, and heart failure
- Monitor ECG in patients with left ventricular hypertrophy or left ventricular dysfunction

Elderly

- Some patients may tolerate lower doses better



Children and Adolescents

- Safety and efficacy have not been established
- Use with caution, observing for activation of known or unknown bipolar disorder and/or suicidal ideation, and strongly consider informing parents or guardian of this risk so they can help observe child or adolescent patients



Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- Some animal studies have shown adverse effects

Breast Feeding

- Unknown if dextromethorphan/quinidine is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- If child becomes irritable or sedated, breast feeding or drug may need to be discontinued

- Must weigh benefits of breast feeding with risks and benefits of antidepressant treatment versus nontreatment to both the infant and the mother

The art of psychopharmacology

Potential Advantages

- No other approved agent for PBA

Potential Disadvantages

- CYP450 2D6 poor metabolizers may require dose reduction
- In patients with past substance abuse, especially of dextromethorphan, ketamine, or PCP

Primary Target Symptoms

- Uncontrollable crying
- Uncontrollable laughter



Pearls

- Quinidine is intended to increase the actions of dextromethorphan by inhibiting its metabolism by CYP450 2D6; therefore, poor metabolizers of CYP450 2D6 may not benefit as much from this treatment while still experiencing adverse effects associated with quinidine

- Affective instability in Alzheimer disease may be treatable with this agent, including agitation, allowing antipsychotics to be avoided in this population
- Some men express emotional lability as laughter and anger rather than laughter and crying
- Affective instability in PTSD and in mild traumatic brain injury may be improved by dextromethorphan/quinidine
- Similar binding properties to ketamine suggest possible efficacy in both treatment-resistant depression and chronic pain

Suggested Reading

Brooks BR , Thisted RA , Appel SH , et al. Treatment of pseudobulbar affect in ALS with dextromethorphan/quinidine: a randomized trial . *Neurology* 2004 ;**63** (8):1364 –70.

Garnock-Jones KP . Dextromethorphan/quinidine in pseudobulbar affect . *CNS Drugs* 2011 ;**25** (5):435–45 .

The Medical Letter Inc . Dextromethorphan/quinidine (neudexta) for pseudobulbar affect . *Med Lett Drugs Ther* 2011 ;**53** (1366):46 –7.

Pioro EP , Brooks BR , Cummings J , et al. Dextromethorphan plus ultra low-dose quinidine reduces pseudobulbar affect . *Ann Neurol* 2010 ;**68** (5):693–702 .

Diazepam

Therapeutics

Brands

- Valium
- Diastat
- *see index for additional brand names*

Generic?

Yes (not Diastat)

Class

- Neuroscience-based Nomenclature: GABA positive allosteric modulator (GABA-PAM)
- Benzodiazepine (anxiolytic, muscle relaxant, anticonvulsant)

Commonly Prescribed for

(bold for FDA approved)

- **Anxiety disorder**
- **Symptoms of anxiety (short-term)**
- **Acute agitation, tremor, impending or acute delirium tremens and hallucinosis in acute alcohol withdrawal**

- **Skeletal muscle spasm due to reflex spasm to local pathology**
- **Spasticity caused by upper motor neuron disorder**
- **Athetosis**
- **Stiff-person syndrome**
- **Convulsive disorder (adjunctive)**
- **Anxiety during endoscopic procedures (adjunctive) (injection only)**
- **Preoperative anxiety (injection only)**
- **Anxiety relief prior to cardioversion (intravenous)**
- **Initial treatment of status epilepticus (injection only)**
- **Insomnia**
- **Catatonia**

How the Drug Works

- Binds to benzodiazepine receptors at the GABA-A ligand-gated chloride channel complex
- Enhances the inhibitory effects of GABA
- Boosts chloride conductance through GABA-regulated channels
- Inhibits neuronal activity presumably in amygdala-centered fear circuits to provide therapeutic benefits in anxiety disorders
- Inhibiting actions in cerebral cortex may provide therapeutic benefits in seizure disorders

- Inhibitory actions in spinal cord may provide therapeutic benefits for muscle spasms

How Long Until It Works

- Some immediate relief with first dosing is common; can take several weeks with daily dosing for maximal therapeutic benefit

If It Works

- For short-term symptoms of anxiety or muscle spasms –after a few weeks, discontinue use or use on an “as-needed” basis
- Chronic muscle spasms may require chronic diazepam treatment
- For chronic anxiety disorders, the goal of treatment is complete remission of symptoms as well as prevention of future relapses
- For chronic anxiety disorders, treatment most often reduces or even eliminates symptoms, but not a cure since symptoms can recur after medicine stopped
- For long-term symptoms of anxiety, consider switching to an SSRI or SNRI for long-term maintenance
- If long-term maintenance with a benzodiazepine is necessary, continue treatment for 6 months after symptoms resolve, and then taper dose slowly
- If symptoms reemerge, consider treatment with an SSRI or SNRI, or consider restarting the benzodiazepine; sometimes benzodiazepines have to be used in combination with SSRIs or SNRIs for best results

If It Doesn't Work

- Consider switching to another agent or adding an appropriate augmenting agent
- Consider psychotherapy, especially cognitive behavioral psychotherapy
- Consider presence of concomitant substance abuse
- Consider presence of diazepam abuse
- Consider another diagnosis, such as a comorbid medical condition



Best Augmenting Combos for Partial Response or Treatment Resistance

- Benzodiazepines are frequently used as augmenting agents for antipsychotics and mood stabilizers in the treatment of psychotic and bipolar disorders
- Benzodiazepines are frequently used as augmenting agents for SSRIs and SNRIs in the treatment of anxiety disorders
- Not generally rational to combine with other benzodiazepines
- Caution if using as an anxiolytic concomitantly with other sedative hypnotics for sleep

Tests

- In patients with seizure disorders, concomitant medical illness, and/or those with multiple concomitant long-term medications, periodic liver tests and blood counts may be prudent